

2. Common Elements Between Two Arrays and Complexity

Aim

To find the common elements between two arrays and analyze the time and space complexity of the algorithm.

Algorithm

1. Read two arrays.
2. Compare every element of the first array with every element of the second array.
3. If an element is present in both arrays, add it to the result.
4. Display the common elements.

Procedure

1. Start the program.
2. Read the size and elements of the first array.
3. Read the size and elements of the second array.
4. Use nested loops to compare the elements.
5. If two elements are equal, store the element.
6. Display the common elements.
7. Stop the program.

main.py	Visualize	Run ▶	Output
<pre>1 n = int(input("Enter size of first array: ")) 2 a = list(map(int, input("Enter elements: ").split())) 3 4 m = int(input("Enter size of second array: ")) 5 b = list(map(int, input("Enter elements: ").split())) 6 7 common = [] 8 9 for i in a: 10 for j in b: 11 if i == j and i not in common: 12 common.append(i) 13 14 print("Common Elements:", common)</pre>			<pre>Enter size of first array: 3 Enter elements: 1 5 3 Enter size of second array: 3 Enter elements: 1 5 4 Common Elements: [1, 5]</pre>

Result

Thus, the common elements between the two given arrays were successfully identified and their complexity was analyzed.